

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No.- 02

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Recap of Previous Lecture



Topic

Pipeline Processing



Topics to be Covered



Topic

Pipeline Processing

Topic

Instruction Pipeline

- #Q. The time delay of the four segments in pipeline are as shown follows:
 $t_1 = 50 \text{ ns}$, $t_2 = 30 \text{ ns}$, $t_3 = 95 \text{ ns}$, and $t_4 = 45 \text{ ns}$.
The interface registers delay time $t_r = 5 \text{ ns}$.
How long would it take to process 100 tasks in the pipeline?

$$\begin{aligned} \text{Processing time} &= (4 + 100 - 1) 100 \\ &= 10300 \text{ ns} \end{aligned}$$

$$t_n = 220 \text{ ns}$$

$$S = \frac{220 * 100}{10300}$$

$$= 2.14$$

#Q. How can we reduce the total time about the one-half of the time calculated in above question?

if seg. 3 is divided into 2 segments $\Rightarrow$ $\begin{matrix} \rightarrow S_{31} & 50 \text{ ns} \\ \rightarrow S_{32} & 45 \text{ ns} \end{matrix}$

$$\left. \begin{array}{l} S_1 \Rightarrow 50 \text{ ns} \\ S_2 \Rightarrow 30 \text{ ns} \\ S_{31} \Rightarrow 50 \text{ ns} \\ S_{32} \Rightarrow 45 \text{ ns} \\ S_4 \Rightarrow 45 \text{ ns} \end{array} \right\} k=1$$

$$t_p = 50 + 5 = 55 \text{ ns}$$

$$\begin{aligned} \text{Time} &= (5 + 100 - 1) 55 \\ &= 5720 \end{aligned}$$

$$\begin{aligned} S &= \frac{220 * 100}{5720} \\ &= 3.85 \end{aligned}$$

ques) $k = 5$ segments
all segments take delay $\Rightarrow 4\text{ns}$

$$\left. \begin{array}{l} t_p = 4\text{ns} \\ t_n = 20\text{ns} \end{array} \right\} t_n = k * t_p$$

if all segments take equal
delay and there is no any
buffer or reg. used.

$$\Downarrow \\ \text{then } t_n = k t_p$$

$$\Downarrow \\ S_{\text{ideal}} = k$$

#Q. Consider a non-pipelined processor with a clock rate of 8 gigahertz and average cycles per instruction of 6. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 6 gigahertz. Assume that there are no stalls in the pipeline. The speed up achieved in this pipelined processor is 4.5.

$$\text{non-pipeline:-}$$
$$\text{cycle time} = \frac{1}{8\text{GHz}}$$

$$t_n = 6 * \frac{1}{8\text{GHz}} = 0.75\text{ns}$$

$$\text{pipeline:-}$$

$$\text{cycle time } (t_p) = \frac{1}{6\text{GHz}} = 0.167\text{ns}$$

$$S = \frac{0.75}{0.167} = 4.5$$



Topic : Latency and Throughput

after how much time a new input is given

latency of pipeline = t_p

— || — non-pipeline = t_n

no. of operations performed per unit time

in $(k+n-1)t_p$ time no. of operations = n

$$\text{in } 1 \text{ ———— } = \frac{n}{(k+n-1)t_p}$$

in ideal condition throughput = $\frac{1}{t_p}$

Throughput of non pipeline = $\frac{1}{t_n}$

#Q. Consider 6 segment pipeline with segment delay of segments as 120ns, 126ns, 121ns, 110ns, 118ns and 120ns respectively. The intermediate buffer delay is 5ns.

Consider that the system is used for performing 100 tasks.

1. What is the latency of non-pipeline system $\Rightarrow 715 \text{ ns}$
2. What is the latency of pipeline system $\Rightarrow 131 \text{ ns}$
3. What is the throughput of pipeline system $\frac{100}{(6+100-1)131 \text{ ns}} = 7270083.6 \text{ per sec}$
4. What is the throughput of pipeline system in ideal case $\frac{1}{131 \text{ ns}} = 7633587.79 \text{ per sec}$

$$t_p = 126 \text{ ns} + 5 = 131 \text{ ns}$$

$$t_n = 715 \text{ ns}$$

#Q. The stage delays in a 4-stage pipeline are 60ns, 50ns, 55ns and 80ns. The last stage (with delay 80ns) is replaced with a functionally equivalent design involving two stages with respective delays 50ns and 30ns. The throughput increase of the pipeline is _____ percent?



#Q. Consider a 6-stage pipeline with delays 2, 4, 3, 5, 3 and 4 ns. This pipeline is upgraded to a new 8-segment pipeline in which each segment delay is 2 ns.

1. How much time is saved using new pipeline over old one for 100 tasks? $\Rightarrow 311$
2. What is the speed up of new pipeline as compared to old pipeline for 100 tasks?

$$\begin{aligned}
 k &= 6 \\
 t_p &= 5 \text{ ns} \\
 \text{time} &= (6 + 100 - 1) 5 \\
 &= 525 \text{ ns}
 \end{aligned}$$

$$\begin{aligned}
 k &= 8 \\
 t_p &= 2 \text{ ns} \\
 \text{time} &= (8 + 100 - 1) 2 \\
 &= 214 \text{ ns}
 \end{aligned}$$

$$\begin{aligned}
 1. \text{ Time Saved} &= \frac{525}{214} \\
 &\quad \downarrow \\
 &= 2.45
 \end{aligned}$$

$$\begin{aligned}
 2. \text{ Speed up} &= \frac{525}{214} \\
 &= 2.45
 \end{aligned}$$

Ques) In prev. questⁿ in ideal condition speed up of new pipeline?

Solⁿ

$$S = \frac{t_{p_1}}{t_{p_2}}$$

$$= \frac{5ns}{2ns}$$

$$= 2.5$$



Topic : Instruction Pipeline



If pipeline processing is implemented on Instruction Cycle





Topic : Instruction Pipeline

Assume following 5-stage pipeline:

- IF: Instruction Fetch
- ID: Instruction Decode & Address Calculation
- OF: Operand Fetch
- EX: Execution
- WB: Write Back



Topic : Instruction Pipeline

no. of inst^{ns} executed = 6 (I1 to I4, I8 to I9)

cycles for 6 inst^{ns} = $5 + 6 - 1 = 10$

stalls = 3

Total = 13

Clock Cycles		1	2	3	4	5	6	7	8	9	10	11	12	13	14
Instructions	I1	IF	ID	OF	EX	WB									
	I2		IF	ID	OF	EX	WB								
	I3			IF	ID	OF	EX	WB							
	I4				IF	ID	OF	EX	WB						
	I5					IF	-	-							
	I6														
	I7														
	I8								IF	ID	OF	EX	WB		
	I9									IF	ID	OF	EX	WB	

CPU checks condition and gets to know that jump to be taken or not.

branch

stall or bubble

Target



Topic : Instruction Pipeline

$$\begin{aligned} \text{no. of cycles} &= 5 + 9 - 1 = 13 \\ \text{stalls} &= 3 \\ \hline \text{Total} &= 16 \end{aligned}$$

Instructions

Clock Cycles

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
I1	IF	ID	OF	EX	WB											
I2		IF	ID	OF	EX	WB										
I3			IF	ID	OF	EX	WB									
I4				IF	ID	OF	EX	WB								
I5					IF	-	-	IF	ID	OF	EX	WB				
I6									IF	ID	OF	EX	WB			
I7										IF	ID	OF	EX	WB		
I8											IF	ID	OF	EX	WB	
I9												IF	ID	OF	EX	WB

false

branch

stall





Topic : Instruction Pipeline

if stage no i provide branch condition, then

$$\text{no. of stalls due to each branch inst}^n = i - 1$$

next inst^n after branch inst^n is fetched after complete execution of branch inst^n .

$$\text{no. of stalls} = k - 1$$



#Q. Consider a program which contains 50 instructions I1, I2 , I3 I50. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I7 and its target is instruction I46. If during the execution of the program the branch is taken then number of cycles required to execute this program in the given pipeline is 19 ?

$$\begin{array}{rcl}
 \text{no. of inst}^{\text{ns}} \text{ executed} & = & 7 \quad (\text{I1 to I7}) \\
 & & 5 \quad (\text{I46 to I50}) \quad (50 - 46 + 1) \\
 \hline
 & & 12
 \end{array}$$

$$\begin{array}{rcl}
 \text{w/o stalls} & = & 5 + 12 - 1 = 16 \\
 \text{stalls for 1 branch inst}^{\text{n}} & = & 4 - 1 = 3 \\
 \hline
 \text{Total} & = & 19
 \end{array}$$

#Q. In the above question if the pipeline has cycle time 10ns then total time required to execute the program is 190 ns?

$$= 19 * 10 \text{ ns}$$

$$= 190 \text{ ns}$$



#Q. Consider a program which contains 200 instructions $I_1, I_2, I_3, \dots, I_{200}$. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The stage delays are 3ns, 4ns, 2ns, 5ns and 4ns respectively. The program contains only one branch instruction which is instruction I_4 and its target is instruction I_{24} . If during the execution of the program the branch is taken then the time required to execute this program in the given pipeline is _____?

$$\begin{aligned}
 \text{no. of instns executed} &= 4 \quad (I_1 \text{ to } I_4) \\
 &= 177 \quad (I_{24} \text{ to } I_{200}) \\
 \hline
 &181 \quad 200 - 24 + 1
 \end{aligned}$$

$$\begin{aligned}
 \text{no. of cycles w/o stall} &= 5 + 181 - 1 = 185 \\
 \text{no. of stalls} &= 4 - 1 = 3 \\
 \hline
 &188
 \end{aligned}$$

$$t_p = 5 \text{ ns}$$

$$\begin{aligned}
 \text{time} &= 188 * 5 \\
 &= 940 \text{ ns}
 \end{aligned}$$

#Q. Consider a program which contains 200 instructions I1, I2 , I3 I200. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. Branch instructions are not overlapped. The program contains only 1 branch instruction which is instruction I8 and its target is instruction I123. If during the execution of the program the branch is not taken then number of cycles required to execute this program in the given pipeline is 207?

no. of instrs executed = 200

$$\begin{aligned} \text{no. of cycles w/o stalls} &= 5 + 200 - 1 = 204 \\ \text{stalls} &= 4 - 1 = 3 \\ \hline \text{Total} &= \underline{\underline{207}} \text{ Ans.} \end{aligned}$$

#Q. Consider a program which contains 50 instructions I1, I2 , I3 I50. Further consider a 5-stage pipeline in which branch condition is evaluated in 3rd stage of pipeline. The program contains only 1 branch instruction which is instruction I20, and its target is instruction I12. If during the execution of the program the branch is taken exactly once, then number of cycles required to execute this program in the given pipeline is 67?

$$\begin{aligned}
 \text{no. of inst}^{\text{ns}} \text{ executed} &= 20 \\
 &= 50 - 12 + 1 \\
 \hline
 \text{Total} &= 59
 \end{aligned}$$

(I1 to I20)
(I12 to I50)

$$\begin{aligned}
 \text{no. of cycles w/o branch} &= 5 + 59 - 1 = 63 \\
 \text{Stalls due to branch} &= (3 - 1) \times 2 = 4
 \end{aligned}$$

67



Topic : Pipeline Hazards



Situations that prevent the next instruction from being executing during its designated clock cycle





Topic : Pipeline Hazards

1. Structural Hazard / Resource Conflict
2. Data Hazard / Data Dependency
3. Control Hazard / Branch Difficulty





Topic : Structural Hazard

2 different instructions try to use same hardware or resource at same time.

ex:

ADD — 1 cycle in EX

MUL — 2 cycles in EX

MUL IF ID OF EX EX WB

ADD

IF ID OF — EX WB

no. solutⁿ $\Rightarrow$ stalls needed



2 mins Summary



Topic

Pipeline Processing

Topic

Instruction Pipeline





Happy Learning

THANK - YOU

